

“Sketch Dance Stage Online”: Three-dimensional CG projection of hand-drawn characters to real space and interaction

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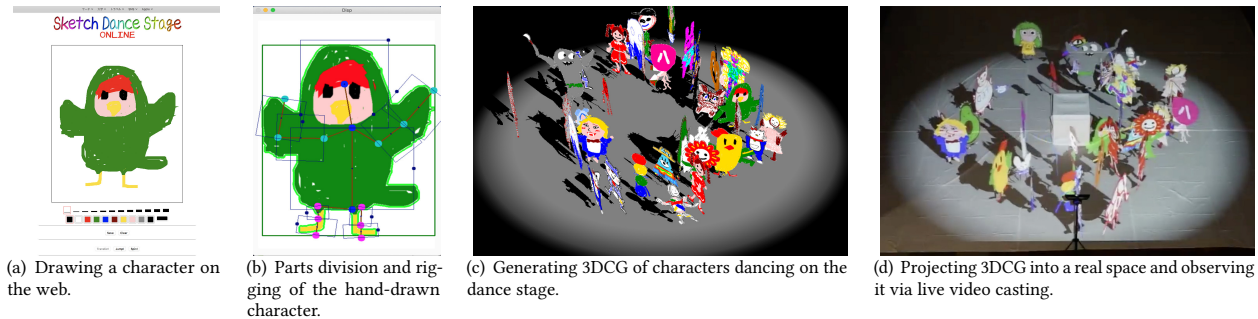


Figure 1: Overview of “Sketch Dance Stage Online”.

KEYWORDS

Sketch, Online, Interaction, 3D Projection Mapping, Occlusion

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1 INTRODUCTION

We introduce an interactive content “Sketch Dance Stage Online” (Fig. 1). In this content, when the user draws a character on a web page, a CG character is generated from the hand-drawn character and appears three-dimensionally in the real space while dancing. The user can also interact with the CG character in several ways.

This study is an extended version of our previous study, “Sketch Dance Stage” [Mizuno et al. 2015]. In the previous study, the user draws a character on a paper with pens, and the system scans it with a scanner or camera. The scanned character is divided into 10 parts such as the head, torso and limbs, and bones are set. As the result, the hand-drawn character is converted into a CG character. The user can give motion to the CG character by his/hers own dance. The CG character given the motion appears on the dance stage in the CG space while dancing, and the user observes it on the monitor. This content has been used at various events.

Due to the COVID-19 pandemic, it is difficult to hold physical events now. Therefore, we make our “Sketch Dance Stage” online. In the online version, the user draws a character on a web page instead of drawing a character on a paper. It is easy to change physical drawing to online drawing using web pages. However, in this case, the content is completed in the digital / virtual space, and there are few contacts with the analog / physical space.

We have also developed “Amazing Sketchbook” as a content that can generate 3DCG by drawing a sketch on a paper with pens [Kondo et al. 2013] [Mizuno and Funahashi 2017]. “Sketch Dance Stage” and “Amazing Sketchbook” are especially popular with children and have been experienced by more than 60,000 people. We think that one of the reasons for the popularity of these contents is the point of contact between the extremely analog operations and tools of drawing a sketch on a paper with pens, and the digital process of generating 3DCG. Therefore, the attractiveness of these sketch-based contents might be lost only by simply changing drawing way physical to online.

In our “Sketch Dance Stage Online”, we focus on the way of showing 3DCG as a new contact point between the digital / virtual space and the analog / physical space. In this content, 3DCG with CG characters generated from hand-drawn sketches is projected on a floor of the real space. Then when the user observes it via YouTube Live or Zoom, the CG characters appear to exist three-dimensionally in the remote real space. At this time, the CG characters fuse with the real space by realizing the correct occlusion with physical objects in the real space.

The user can interact with CG characters that appear in the remote real space via the web page such as to jump or to spin. The user can also observe the interaction between the real space and the CG character, such as the CG character moving in the same way as a person in the remote real space.

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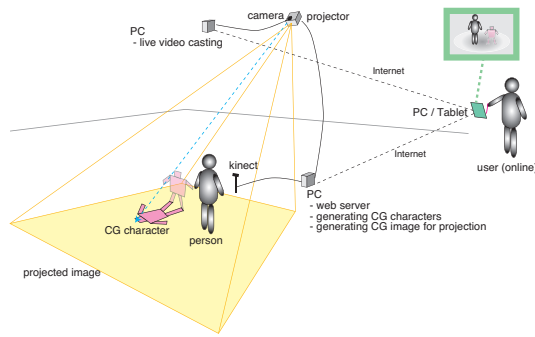


Figure 2: The system for “Sketch Dance Stage Online”.

2 METHODS

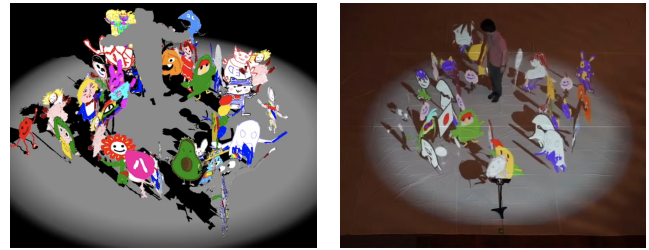
The system for “Sketch Dance Stage Online” consists of two PCs, a projector, a camera, and a Kinect sensor as shown in Figure 2. One PC is used for a web server, generating CG characters from sketches of characters, and generating 3DCG for projection. The other PC is used for live video casting.

The web server runs the web pages for drawing a character and for interacting with a CG character generated from the hand-drawn character. The web page consists of a canvas for drawing, and buttons of “Jump” and “Spin” for interaction (Fig. 1(a)). The web page is implemented in HTML5, PHP and Javascript. The user accesses the web page with a PC or a tablet, and draws a character on the canvas with pens of various colors and thicknesses. Then, the drawn character is sent to the web server as image data with the user’s ID. When the user presses the jump button or the spin button to interact with the CG character, the user’s ID is sent to the web server as motion flags.

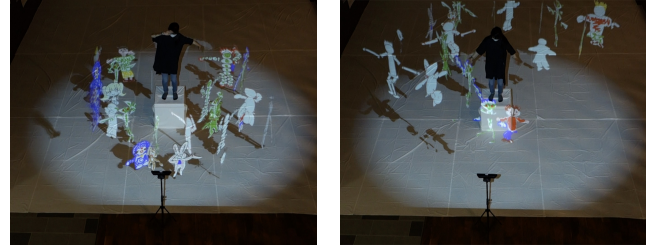
The method of generating a CG character from the character image is the same as the method used in the former “Sketch Dance Stage”. Dividing the character image into parts and rigging are performed automatically but simple, and the operator manually modifies them as needed (Fig. 1(b)). The motion data of the CG character is randomly assigned from the data group acquired in advance.

The method of generating 3DCG image for projection is basically the same as the method used in the former “Sketch Dance Stage”, and it is implemented in C++ with OpenGL. In this content, the 3DCG is generated by reproducing the positional relationship between the floor and the projector in the real space. The CG viewpoint is set based on the position and the optical axis of the projector, and projection transformation is performed based on the viewing angle of the projector (Fig. 1(c)). Then, when the generated 3DCG image is projected on the floor with the projector, the projected image has the effect of anamorphosis, and the CG image is observed three-dimensionally from the position of the projector. Therefore, by placing a camera near the projector, the CG characters are also observed to exist three-dimensionally in the real space in the live video by the camera (Fig. 1(d)).

In our method, conflict of occlusion between physical objects and the projected image may occur since the image is projected to all places visible from the projector. To prevent this problem, the system scans physical objects in the area three-dimensionally where the image is projected using the Kinect sensor installed in



(a) A 3D model of a person generated from the scan data and placed in the 3DCG space with CG characters. (b) Appropriate occlusion between CG characters and a person in the real space.



(c) CG characters performing the same actions as a person in the real space (dancing). (d) CG characters performing the same actions as a person in the real space (jumping).

Figure 3: Fusion of CG characters and a person in the real space.

the real space. Then, 3D models of the physical objects with the same color as the background is generated from the scan data, and placed in the 3DCG space with CG characters for generating an image for projection (Fig. 3(a)). Therefore, when a person stands in the projection area, the CG characters determined to be behind the person are occluded by the scan data of the person, and are deleted from the image. When the image is projected, it is not projected on the person and observed as if the person is standing in front of the CG characters in the live video (Fig. 3(b)).

The Kinect sensor is also used to acquire the motion of a person in the real space. Then, the CG characters can be made to perform the same motion as the person in real space (Fig. 3(c)(d)).

The system constantly checks if motion flags sent by the online users are set. When a motion flag is set, the CG character drawn by the corresponding user is jumped or spun. As the result, online users can also interact with CG characters that appear in the remote real space.

3 EXHIBITION

We exhibited “Sketch Dance Stage Online” from Japan at an online event for people who live in London. About 100 people enjoyed our content online, and we confirmed that the proposed method is useful for online interactive contents.

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